Statistical Mechanics, Lecture notes

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Contents

Lesson 1, 23/09 **4**

Thermodynamics **4**

Lesson 2, 29/09 **4**

1 Internal Energy as a TD potential **4**

2 Legendre Transform **4**

3 Response Functions **5**

4 Mechanical response functions **6**

5 Maxwell Relations **6**

6 Stability of equilibrium state **7**

Lesson 3, 30/09 **7**

7 Equilibrium in external field **8**

8 Second order equilibrium **8**

9 Multi-component systems **10**

10 Phase transitions **10**
10.1 First order transitions 11

Lesson 4, 09/10 **11**
10.2 Continuous transitions 13

11 Maxwell construction **14**

Lesson 5, 07/10 **14**

12 Coexistence of many phases **15**

13 Gibbs rule **15**

Statistical Mechanics **16**

14 Statistical approach to TD **16**

15 Observables **16**

Lesson 6, 13/10 **17**

16 Description of TD systems through independent subsystems **17**
16.1 Additive observables 18
16.2 Explicit formula for ρ 18

17 The Time reversal problem **18**
17.1 Reduced Observables 18

Lesson 7, 14/10 **21**
17.2 Time scales in BBGKY 21

Lesson 8, 20/10 **21**

18 Boltzmann H-theorem **21**

19 Maxwell-Boltzmann Distribution **22**
19.1 About the H-theorem 23

20 Going Quantum **24**
20.1 Density matrix 24

Lesson 9, 27/10 **25**

21 Gibbs Statistical Ensembles **25**
21.1 Microcanonical Ensemble 26
21.2 Example: Classical ideal gas 27

Lesson 10, 28/10 **27**
21.3 Example: Quantum Mechanics 29

Lesson 1, 23/09

Thermodynamics

VEDI APPUNTI TABLET

Lesson 2, 29/09

1 Internal Energy as a TD potential

[10 min. cfr with notes pt 2 on moodle/Remi] $N = const.$

2 Legendre Transform

What if the control variables are the intensive ones, e.g. system in a thermostat? It is useful to use T instead of S: the mathematical tool used is the *Legendre transform* (the one used to transform $L(q, \dot{q}) \to H(q, p)$, using $p = \frac{\partial L}{\partial \dot{q}}$)

$$E(S, V, N, X) \to F(T, V, N, X) := E(S, V, N, X) - TS$$

where S is found by inverting the relation $T = \frac{\partial E}{\partial S}$. F is the Helmholtz free energy

Using the fundamental equation of TD

$$E = TS - pV + \mu N + yX$$

the function F can also be written as

$$F = -pV + \mu N + yX :$$

differentiating this yields

$$\mathrm{d}F \leq -S\mathrm{d}T - p\mathrm{d}V + \mu\mathrm{d}N + y\mathrm{d}X$$

so keeping fixed T, V, N, X, the equilibrium is at minimal F.

What are the conjugated variables? As always

$$S = -\frac{\partial F}{\partial T}$$
$$p = -\frac{\partial F}{\partial V}$$
$$\mu = \frac{\partial F}{\partial N}$$

This process can be done with every intensive variable e.g. enthalpy:

$$E(S, V, N, X) \underset{p=-\frac{\partial E}{\partial V}}{\longrightarrow} H(S, p, N, X) := E(S, V, N, X) + pV$$

and

$$H = TS + \mu N + yX \underset{\text{d and fund. eq.}}{\longrightarrow} \mathrm{d}H \leq T\mathrm{d}S + v\mathrm{d}p + \mu\mathrm{d}N + y\mathrm{d}X$$

so fixing S, p, N, X, the equilibrium is at minimal H.

Doing this with respect to both couples (T, S) and (p, V) yields the Gibbs free energy:

$$G(T, p, N, X) = E(S, V, N, X) - TS + pV$$

using the fundamental equation yields

$$G = \mu N + yX$$

Note In simple systems (without (y, X)) $G = \mu N$. The differential is

$$\mathrm{d}G \leq -S\mathrm{d}T + V\mathrm{d}p + \mu\mathrm{d}N + y\mathrm{d}X$$

that shows that, fixing T, p, N, X, equilibrium is at minimal G.

The last Legendre Transform of interest is the one that yields the *Grand Canonical potential*:

$$E(S, V, N, X) \underset{(T,S),(\mu,N)}{\longrightarrow} \bar{\omega}(T, V, \mu, X) = E(S, V, N, X) - TS - \mu N$$

The fundamental equation yields

$$\bar{\omega} = -pV + yX$$

and differentiating yields

$$\mathrm{d}\bar{\omega} \leq -S\mathrm{d}T - p\mathrm{d}V - N\mathrm{d}\mu + y\mathrm{d}X$$

so the equilibrium is at the minimum of $\bar{\omega}$.

The pressure is simply

$$p = -\frac{\partial\bar{\omega}}{\partial V} = -\frac{\bar{\omega}}{V}$$

3 Response Functions

These functions describe the reaction of variables when another one changes (typically conjugated variables), are very easy to measure and they are deeply connected to the size of the fluctuations of the system (Fluctuation-dissipation theorem).

Def. Heat capacity

$$C = \frac{\delta Q}{\mathrm{d}T}$$

is the amount of heat to be furnished to change the T; it is not an exact differential, so it depends on the transformation.
Heat capacity at constant (V, N): if $\mathrm{d}V = \mathrm{d}N = \mathrm{d}X = 0 \to \delta W = 0$; by the first principle

$$\delta Q = \mathrm{d}E$$

so

$$C_{V,N,X} = C_V = \frac{\delta Q}{\mathrm{d}T} = \left.\frac{\partial E}{\partial T}\right|_{V,N,X}$$

or, using the definition of entropy

$$\delta Q = T\mathrm{d}S$$

and fixing (V, N, X)

$$\mathrm{d}S = \left.\frac{\partial S}{\partial T}\right|_{V,N,X}\mathrm{d}T$$

so

$$C_V = \frac{T\mathrm{d}S}{\mathrm{d}T} = T\left(\frac{\partial S}{\partial T}\right)_{V,N,X}.$$

A more natural approach springs from the Helmholtz free energy:

$$S = -\left.\frac{\partial F}{\partial S}\right|_{V,N,X} \to C_V = -T\frac{\partial^2 F}{\partial T^2}$$

Heat capacity at constant (p, N, X): the first principle states

$$\delta W = p\mathrm{d}V \to \delta Q = \mathrm{d}E + p\mathrm{d}V$$

so

$$C_p = \frac{\delta Q}{\mathrm{d}T} = \left.\frac{\partial E}{\partial T}\right|_{p,V,X} + p \left.\frac{\partial V}{\partial T}\right|_{p,V,X}$$

but also

$$E = G + TS - pV = G - T\frac{\partial G}{\partial T} - pV$$

so

$$C_p = \frac{\partial}{\partial T}\left(G - T\frac{\partial G}{\partial T} - pV\right) + p\frac{\partial V}{\partial T} = -T\left.\frac{\partial^2 G}{\partial T^2}\right|_{p,N,X}$$

4 Mechanical response functions

Susceptibility χ is the volume response to a change in pressure: this also varies depending on the transformation.

Isothermal: fixing (T, N, X)

$$\chi_T := -\left.\frac{\partial V}{\partial p}\right|_{T,N,X}$$

since the variables are (T, p, N, X), with only p independent, the most natural TD potential to use is G: some calculations show that

$$\chi_T = -\left.\frac{\partial^2 G}{\partial p^2}\right|_{T,N,X}$$

Adiabatic: fixing (S, N, X)

$$\chi_S = -\left.\frac{\partial V}{\partial p}\right|_{S,N,X}$$

the TD potential is the enthalpy H:

$$V = \left.\frac{\partial H}{\partial p}\right|_{S,N,X} \to \chi_S = \left.\frac{\partial^2 H}{\partial p^2}\right|_{S,N,X}$$

All these response functions are between conjugated variables: what happens if they are not? Consider the Thermal expansion coefficient

$$\alpha_{p,N,X} = \alpha_p = \left.\frac{\partial V}{\partial T}\right|_{p,N,X} \underset{\text{exercise}}{=} \left.\frac{\partial^2 G}{\partial p \partial T}\right|_{p,N,X}$$

5 Maxwell Relations

These are relations between TD quantities: consider the set of independent variables

$$\{Z_A\}$$

their conjugated variables

$$\{\xi_A\}$$

and a TD potential

$$\mathrm{d}\Phi = \sum \xi_A \mathrm{d}Z_A$$

it follows that

$$\frac{\partial \xi_A}{\partial Z_B} = \frac{\partial^2 \Phi}{\partial Z_A \partial Z_B} = \frac{\partial^2 \Phi}{\partial Z_B \partial Z_A} = \frac{\partial \xi_B}{\partial Z_A}$$

from other mathematical properties (like the "chain rule", see Billò notes) arise other relations. One finds

$$\frac{1}{T} C_p > \frac{1}{T} C_V + \frac{\alpha_p^2}{\chi_T}$$

6 Stability of equilibrium state

Even at equilibrium, at maximum S, there can be small fluctuations in TD variables. For all such fluctuations, the entropy **must decrease** to ensure stability.

What are the conditions for local equilibrium? Consider an isolated system at equilibrium subdivided in 2 parts, which can exchange particles and heat.

$$E = E_1 + E_2$$
$$V = V_1 + V_2$$
$$N = N_1 + N_2$$

Suppose some fluctuation occurs in E_i, V_i, N_i: the other part of the system must have an opposite fluctuation

$$\Delta E_2 = -\Delta E_1 ...$$

Assuming small fluctuations, the variation of S at first order is

$$\Delta S = \sum_{i=i}^{2} \left(\left.\frac{\partial S_i}{\partial E_i}\right|_{\text{eq}} \Delta E_i + \left.\frac{\partial S_i}{\partial V_i}\right|_{\text{eq}} \Delta V_i + \left.\frac{\partial S_i}{\partial N_i}\right|_{\text{eq}} \Delta N_i \right) = \left(\frac{1}{T_1^{\text{eq}}} - \frac{1}{T_2^{\text{eq}}} \right) \Delta E_1 + \left(\frac{p_1^{\text{eq}}}{T_1^{\text{eq}}} - \frac{p_2^{\text{eq}}}{T_2^{\text{eq}}} \right) + ...$$

Since $\Delta S \leq 0$, it must be that all the quantities are equal

$$T_1^{\text{eq}} = T_2^{\text{eq}} ...$$

Iterating this process for smaller and smaller partitions of the system proves that, unsurprisingly, at first order

$$\Delta S = 0.$$

The chemical potential condition

$$\mu_1^{\text{eq}} = \mu_2^{\text{eq}}$$

guarantees chemical equilibrium: supposing an isolated system with $\delta Q = \delta W = 0$, $\Delta S > 0$ and

$$\Delta S \approx -\left(\frac{\mu_1}{T_1} - \frac{\mu_2}{T} \right) \Delta N_1 > 0$$

so if $\mu_1 > \mu_2$ it must be that

$$\Delta N_1 < 0,$$

in other words, matter flows from higher to lower μ.

The quantity

$$-\boldsymbol{\nabla} \left(\frac{\mu}{T} \right)$$

is a generalized force (see first lesson), much like $\boldsymbol{\nabla} \left(\frac{1}{T} \right)$.

Lesson 3, 30/09

7 Equilibrium in external field

The TD system is often subject to external forces, like gravity: how can equilibrium be described in this case?

Consider a TD system and subdivide it into smaller systems s.t they are still TD systems with a large number of particles: the coordinates of the portion is $\mathbf{x}$.

Since μ is the energy needed to add particles to the system, the control variables used are T, p, N: from the definition of G

$$\mu = \frac{G}{N}$$

and from the fundamental equations

$$\frac{\partial \mu}{\partial p} = \frac{1}{N}\frac{\partial G}{\partial p} := \frac{V(T,p,N)}{N} = v(T,p)$$

where v is the volume per particle.

The chemical potential consists of two parts

$$\mu(T,p;\mathbf{x}) = \mu_0(T,p) + u(\mathbf{x})$$

where μ_0 is μ with the external field turned off and u is related to the field.

At the equilibrium, one expects that T remains constant, p depends on $\mathbf{x}$ (like the Stevino law) and μ remains constant and homogeneous.

These conditions translate to

$$\frac{\partial \mu}{\partial x^i} = 0 = \frac{\partial \mu}{\partial p}\cdot\frac{\partial p}{\partial x^i} + \frac{\partial u}{\partial x^i}$$

so

$$v(T,p)\frac{\partial p}{\partial x^i} + \frac{\partial u}{\partial x^i}.$$

Example: gravity In this case $u(\mathbf{x}) = mgz$, so

$$\frac{\mathrm{d}p}{\mathrm{d}z} v(T,p) = -mg$$

so v does not depend on T, adding the incompressibility of the fluid imposes constant v, so

$$\frac{\mathrm{d}p}{\mathrm{d}z} = -\frac{m}{v}gz = -\rho g z \to p(z) = p_0 - mgz$$

in the case of an ideal gas, using $pv = k_b T$ yields (prove it as an exercise)

$$p(T,z) = p_0 \exp\left[-\frac{mg}{kT}z\right]$$

8 Second order equilibrium

It was found that, at first order

$$T, p, \mu \text{ are homogeneous}$$
$$\Delta S = 0$$

To discuss the second order fluctuations, it is needed to re-express everything in terms of TD potentials: consider the control variables

$$\left\{ \underbrace{X_1, \dots X_r}_{\text{extensive}}; \underbrace{I_{r+1}, \dots I_n}_{\text{intensive}} \right\}$$

and the potential

$$\Phi(X_i; I_j) :$$

its differential is

$$\mathrm{d}\Phi = \sum_{i=1}^{r} I_i \mathrm{d}X_i - \sum_{j=r+1}^{n} X_j \mathrm{d}I_j$$

Note This generalizes the problem to have r extensive control variables and $r - n$ (?) intensive control variables.

Consider a TD system subdivided in two parts and a fluctuation in the extensive variables:

$$\delta X_i = \delta X_{i,1} + \delta X_{i,2} = 0$$

at first order

$$\Delta\Phi^{(1)} = \sum_{i=1}^{r} \left[\left(\frac{\partial \Phi}{\partial X_i} \right)_1 - \left(\frac{\partial \Phi}{\partial X_i} \right)_2 \right] \delta X_{i,1} \geq 0$$

As before, the only way to ensure this condition is

$$\left(\frac{\partial \Phi}{\partial X_i} \right)_1 = \left(\frac{\partial \Phi}{\partial X_i} \right)_2 \iff I_{i,1} = I_{i,2}$$

with finer partitions this means that the intensive variables conjugated to the extensive control variables are homogeneous in the system, at leas at first order.

To get a non-trivial result, consider the second order of fluctuation of a single variable X_i (i.e. considering the fluctuations independent):

$$\Delta\Phi^{(2)} = \frac{1}{2} \left[\left(\frac{\partial^2 \Phi}{\partial X_i^2} \right)_1 + \left(\frac{\partial^2 \Phi}{\partial X_i^2} \right)_2 \right] (\delta X_{i,1})^2 \geq 0$$

where the minus sign was destroyed by the square; this statement is true only if

$$\frac{\partial^2 \Phi}{\partial X_i^2} = \frac{\partial I_i}{\partial X_i} \geq 0$$

from this follows that

$$\frac{\partial X_i}{\partial I_i} \geq 0$$

and, remembering the Legendre transform

$$\Phi(..., X_i, ...) \to \tilde{\Phi}(..., I_i ...) \iff X_i = -\frac{\partial \tilde{\Phi}}{\partial I_i}$$

this gives

$$\frac{\partial^2 \tilde{\Phi}}{\partial I_i^2} \leq 0$$

This gives a geometric condition on the TD potentials at equilibrium: they are either a concave function of intensive variables or convex functions of extensive variables (or vice-versa **CHECK!!!!!!!**) **Example** Consider the Helmholtz free energy: the stability condition implies that

$$\frac{\partial^2 F}{\partial T^2} \leq 0$$

but

$$C_V = -T \frac{\partial^2 F}{\partial T^2} \to C_V \geq 0$$

and also

$$\underbrace{\frac{\partial^2 F}{\partial V^2}}_{\geq 0} = -\frac{\partial p}{\partial V} = -\frac{1}{\frac{\partial V}{\partial p}} = \frac{1}{\chi_T} \geq 0 \iff \chi_T \geq 0$$

Example Consider the Gibbs energy $G(T, p, N)$:

$$\frac{\partial^2 G}{\partial T^2} \leq 0$$

and the C_p definition

$$C_p = -T\frac{\partial^2 G}{\partial T^2} \geq 0.$$

Again the χ_T definition is

$$\chi_T = -\underbrace{\frac{\partial^2 G}{\partial p^2}}_{\leq 0} \geq 0$$

which is consistent with the previous example.

From the relation

$$C_p = C_V + \frac{T\alpha_p^2}{\chi_T}$$

and the stability conditions discussed it follows that, in general

$$C_p \geq C_V$$

9 Multi-component systems

Some TD systems have different components mixed (e.g. the atmosphere), indexed with $a = 1, 2...c$ so the chemical potential is not unique: μ_a, N_a and these potentials depend separately from those variables (which vars???)

$$\mathrm{d}E = T\mathrm{d}S - p\mathrm{d}V + \sum_{a=1}^{c} \mu_a \mathrm{d}N_a + y\mathrm{d}X$$

Each potential has its own equilibrium condition $\mu_{a,1} = \mu_{a,2}$ and intensive variables depend only on $c - 1$ ratios

$$\nu_a = \frac{N_a}{N}$$

with the constraint

$$\sum_{a=1}^{c} N_a = N \iff \sum_{a=1}^{c} \nu_a = 1$$

10 Phase transitions

This topic is very difficult to treat mathematically and one of the most important results of statistical mechanics is the description of this phenomenon.

What are the conditions of coexistence of different phases of matter in equilibrium (e.g. ice and water)? At equilibrium, the state of the TD system is characterized by E, V, N: it is not necessary that at equilibrium state these variables are homogeneous.

Each different phase (each defined by its state equation) of the system has its set of variables p_i, T_i, μ_i.

If the state is equilibrium, the intensive variables must be homogeneous:

$$p_1 = p_2, \quad T_1 = T_2, \quad \mu_1 = \mu_2$$

it is then natural to take p, T as control variables, making μ a dependent variable; since the phases have different equations of state, the potentials will be different: for the phase α

$$G_\alpha(T, p, N_\alpha) = \mu_\alpha N_\alpha \to \mu_\alpha = \frac{G_\alpha}{N_\alpha}.$$

The phases can only coexist on *curves of coexistence* in the state space: for generic values of p, T, $\mu_\alpha \neq \mu_\beta$ and so the phases do not coexist and, considering N constant, the system "selects" the phase which minimizes

$$\mu = \frac{G}{N}.$$

So phases α and β coexist if and only if the values of p, T are so that

$$\mu_\alpha(p, T) = \mu_\beta(p, T)$$

which typically is a curve $p(T)$ in the state space. When this curve is crossed, a phase transition occurs and it can be of two natures.

10.1 First order transitions

The coexistence curve is determined by the intersection of the surfaces $\mu_\alpha(T, p)$ and $\mu_\beta(T, p)$. In general, the value of the derivatives differ at the intersection point(s): fixing p they are

$$\left(\frac{\partial \mu_\alpha}{\partial T}\right)_p = \frac{1}{N_\alpha}\left(\frac{\partial G}{\partial T}\right)_p = -\frac{S_\alpha}{N_\alpha} := -s_\alpha$$

which is the entropy per particle, so in general

$$s_\alpha \neq s_\beta$$

Suppose $s_\alpha > s_\beta$: for $T = T(p) - \varepsilon$, $S_{(-)} = N s_\beta$, while for $T = T(p) + \varepsilon$, $S_{(+)} = N s_\alpha$; in this case the transition from β to α makes the entropy jump of

$$\Delta S = N(s_\alpha - s_\beta) > 0$$

this is nothing but the effect of latent heat, the heat to be furnished to the system just to do a phase transition

$$L = T(p)\Delta S.$$

From an enthalpy p.o.v,

$$H(S, p, N) = G + TS$$

since $\mu_\alpha = \mu_\beta$, $\Delta G = 0$ and so

$$\Delta H = \Delta(TS) = T\Delta S = L.$$

This is similar for fixed temperature transitions, with derivative

$$\left(\frac{\partial \mu}{\partial p}\right)_{T,N} = \frac{V}{N} = v$$

which is the volume per particle; so in general $v_\alpha \neq v_\beta$ and the volume of the system changes during the transition *even at constant pressure* (e.g. ice melting).

Lesson 4, 09/10

[...] qualcosa su T fissata, $p = p(T) \pm \varepsilon$ usando G si vede che c'è un ΔV

How can the shape of the transition curve $p(T)$ be determined? With the *Clausius-Clapeyron equation.*

Recall that

$$\mu(T,p) = \frac{G}{N} \quad \mathrm{d}\mu = -s\mathrm{d}T + v\mathrm{d}p$$

and this has to be satisfied at each point on the curve: consider the infinitesimal change in μ along the curve

$$\mu_\alpha \to \mu_\alpha + \mathrm{d}\mu_\alpha$$
$$\mu_\beta \to \mu_\beta + \mathrm{d}\mu_\beta$$

since $\mu_\alpha = \mu_\beta$ at all points,

$$\mathrm{d}\mu_\alpha = \mathrm{d}\mu_\beta$$

substituting yields

$$-s_\alpha \mathrm{d}T + v_\alpha \mathrm{d}p = -s_\beta \mathrm{d}T + v_\beta \mathrm{d}p$$

which implies

$$\frac{\mathrm{d}p}{\mathrm{d}T} = \frac{\Delta s}{\Delta v} = \frac{\Delta S}{\Delta V}$$

in terms of the latent heat $L = T\Delta S$ it can be written as

$$\frac{\mathrm{d}p}{\mathrm{d}T} = \frac{L}{T\Delta V} = \frac{1}{T}\frac{l}{\Delta v} = \frac{1}{T}\frac{l_{\mathrm{mol}}}{\Delta v_{\mathrm{mol}}} = \frac{1}{T}\frac{l_{\mathrm{sp}}}{\Delta v_{\mathrm{sp}}}$$

where $l = \frac{L}{N}$ is the latent heat per particle, l_{mol} is the molar latent heat and l_{sp} is the specific latent heat (J/kg).

Example water to ice transition
In standard conditions, $p_{\mathrm{atm}} = 101325$ Pa, $T_0 = 273.15$ K, the transition is possible and, contrary to many substances, water is denser than ice: the variation of specific volume for the transition is

$$\Delta v_{\mathrm{sp}} = v_{\mathrm{sp}}^{\mathrm{ice}} - v_{\mathrm{sp}}^{\mathrm{water}} = 9 \cdot 10^{-5} \ \mathrm{m}^3/\mathrm{kg}$$

and the specific latent heat is

$$l_{\mathrm{sp}} = 2.3 \cdot 10^5 \ \mathrm{J/kg}$$

The question is, how much pressure is needed to melt the ice at temperature $T_0 - 1$ K?

In this case, $1 \ \mathrm{K} = \Delta T << T_0$ and the C-C equation becomes

$$\frac{\mathrm{d}p}{\mathrm{d}T} = \frac{1}{T}\frac{l_{\mathrm{sp}}}{\Delta v_{\mathrm{sp}}} \to \frac{\Delta p}{\Delta T} = \frac{1}{T}\frac{l_{\mathrm{sp}}}{v_{\mathrm{sp}}} \to \Delta p = \frac{\Delta T}{T_0}\frac{l_{\mathrm{sp}}}{\Delta v_{\mathrm{sp}}} \approx 10^7 \ \mathrm{Pa}$$

Example vapour at saturation
When the liquid and gas phases coexist, the vapour is at saturation; this is a first order transition in which $l > 0$ and $v_{\mathrm{gas}} >> v_{\mathrm{liquid}}$, so $\Delta v \approx v_{\mathrm{gas}}$. The C-C equations takes form as

$$\frac{\mathrm{d}p}{\mathrm{d}T} = \frac{l}{T v_{\mathrm{gas}}}$$

assuming a perfect gas that satisfies the ideal state equation

$$pV = NkT \to p v_{\mathrm{gas}} = kT$$

the equation becomes

$$\frac{\mathrm{d}p}{\mathrm{d}T} = \frac{l}{T}\frac{p}{kT} \to \frac{\mathrm{d}p}{p} = \frac{l}{k}\frac{\mathrm{d}T}{T^2} \to \ln\left(\frac{p}{p_0}\right) = -\frac{l}{k}\left(\frac{1}{T} - \frac{1}{T_0}\right)$$

so the form of the transition curve is an exponential

$$p(T) = p_0 \exp\left[-\frac{l}{k}\left(\frac{1}{T} - \frac{1}{T_0}\right)\right]$$

What about the pV plane? Since the control variables can always be chosen, it should not be fundamentally different: consider a coexistence curve with a point $(p(T_0), T_0)$ and $\frac{\mathrm{d}p}{\mathrm{d}T} > 0$ and a transition at constant T and decreasing p; the transition follows an isothermal curve in the pV plane, then at $p(T_0)$ it becomes "isobara" while the volume changes, this is the coexistence curve, and finally it becomes isothermal again. Consider the coexistence curve: during the "isobara"

$$V = N_\alpha v_\alpha \underset{\text{"isobara"}}{\longrightarrow} V = N_\alpha v_\alpha + N_\beta v_\beta \to v = \nu_\alpha v_\alpha + \nu_\beta v_\beta$$

where $\nu_i = N_i/N$; from this follows

$$v - v_\alpha = \nu_\beta(v_\beta - v_\alpha)$$

and

$$v_\beta - v = \nu_\alpha(v_\beta - v_\alpha)$$

their quotient yields

$$\frac{v - v_\alpha}{v_\beta - v} = \frac{\nu_\beta}{\nu_\alpha}$$

Considering all the different values of T shows that in the pV plane the coexistence happens in a 2D region instead of a curve.

v_α and v_β are implicitly determined by

$$\begin{cases} p_\alpha = p_\beta \\ \mu_\alpha = \mu_\beta \end{cases}$$

In terms of free energy per particle $f = F/N$, $p = -\frac{\partial f}{\partial v}$: since p is constant in the phase transition, $f(v)$ is linearly decreasing with slope

$$-p = \frac{f_\alpha - f_\beta}{v_\beta - v_\alpha}$$

so

$$f_\beta - f_\alpha = -p(v_\beta - v_\alpha) \to (f + pv)_\alpha = (f + pv)_\beta$$

recalling that F is related with a Legendre transform to the Gibbs free energy G,

$$f + pv = G/N = \mu$$

it implies

$$\mu_\alpha = \mu_\beta.$$

It can be shown that

$$f(v) = \nu_\alpha f_\alpha + \nu_\beta f_\beta$$

10.2 Continuous transitions

In this case the first derivatives of μ are continuous and that means

$$\Delta S = 0 \quad \Delta V = 0$$

and the discontinuity is at higher orders. Typically the coexistence in the plane p, T is represented by single points, as opposed to curves for first order transitions: these are called *critical points* and usually they are endpoints of first order transitions.

11 Maxwell construction

Suppose to know an approximated empirical equation of state for some system

$$p = p(V, T)$$

that in some regions *does not* obey the stability conditions, i.e. in some regions, at constant temperature,

$$\left.\frac{\partial p}{\partial V}\right|_{T,N} > 0$$

this is not possible and means that the equation of state is wrong, but suppose it works well in other regions: how can this be corrected?
There is a method to connect "good" regions with one another, due to Maxwell, that consists in supposing that the "bad" region represents a phase transition: for example connecting two points in the pV plane with a straight line at $p = \bar{p}$; this connection must be between two points V_1, V_2 so that the free energy at those points is the same as the one determined by the equation of state.

$$\mathrm{d}F = -S\mathrm{d}T - p\mathrm{d}V + \mu\mathrm{d}N$$

fixing T, N this becomes

$$F(V_2, T, N) - F(V_1, T, N) = -\int_{V_1}^{V_2} \mathrm{d}V p(V, T, N)$$

On the contrary, for the "isobara" the difference in free energy is

$$F(V_2, T, N) - F(V_1, T, N) = \bar{p}(V_2 - V_1)$$

so the condition is

$$\bar{p}(V_2 - V_1) = \int_{V_1}^{V_2} \mathrm{d}V p(V, T, N)$$

Example: Van Deer Waals
The VdW equation considers intermolecular interactions with a potential "wall" at distance d and for $r > d$ a $\sim -\frac{1}{r}$ potential: this is a good approximation of the real potential that governs intermolecular interactions; this also determines an effective volume

$$V_{\text{eff}} = V - b$$

as the single molecule's "volume" is $\sim d^3$ and this volume is unavailable for the rest of the system. The well after distance d forms bound states, which lowers the kinetic pressure:

$$p = p_{\text{kin}} - \frac{a}{V^2}$$

this defect in pressure is proportional to V^{-2} (why?).

The last hypothesis is that the kinetic pressure and the effective volume follow the ideal gas law

$$p_{\text{kin}} V_{\text{eff}} = nRT$$

substituting yields the VdW equation of state

$$\left(p + \frac{a}{V^2}\right)(V - b) = nRT$$

since $p(V)$ is approximately dependent on V^{-3}, some isothermals are non-physical, since they present horizontal flex points.

Lesson 5, 07/10

These non-physical isothermals appear under a critical temperature T_c, determined by the parameters a, b.

[equation of correspondent states]

12 Coexistence of many phases

Consider a p, V, T system with 1 type of component: at most three phases can coexist, because the equality of the chemical potentials depends only on 2 variables p, T, so only 2 conditions can have a single solution

$$\begin{cases} \mu_\alpha(p,T) = \mu_\beta(p,T) \\ \mu_\beta(p,T) = \mu_\gamma(p,T) \end{cases}.$$

The 3 phases can coexist at most in single points, called *triple points*.

Example: triple point of water

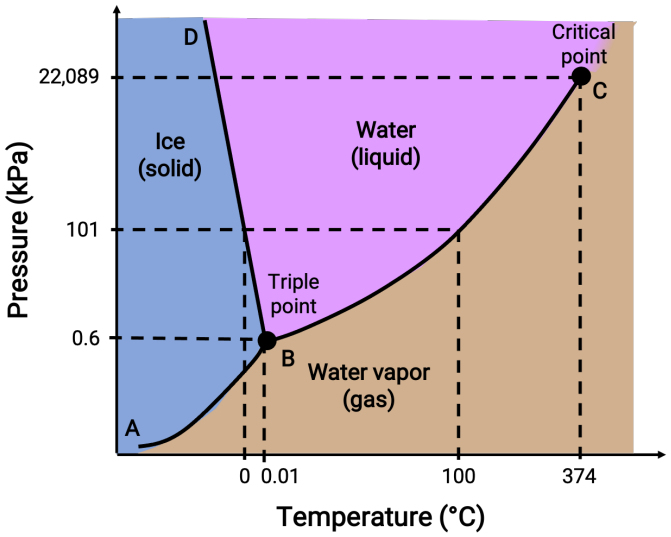


Fig. 12.1: The phase diagram of water.

The liquid-gas transition is continuous, but the coexistence line has an end, this means that their distinction is well-defined only "close" to the first order transition and that they are not so different from each other (they share the same symmetry in structure); at the critical point there usually is a partial symmetry breaking.

13 Gibbs rule

This is a statement about the number of phases that can coexist in multi-component systems.
Consider a system with c components (indexed by $i = 1, 2..c$), which presents f phases at equilibrium (indexed by $\alpha = 1, 2...f$); the portion of particles of type i in phase α is denoted by $\nu_{i,\alpha}$. For all α, it is true that

$$\sum_{i=1}^{c} \nu_{i,\alpha} = 1$$

so these are $f \cdot (c-1)$ constraints.
Moreover, for every component i the chemical potentials for each phase must be the same

$$\mu_{i,1}(T, p, \nu_{1,1}, ...\nu_{c-1,1}) = \mu_{i,2}(...) = \mu_{i,3} = ... = \mu_{i,f}$$

these are $f-1$ independent equations for every i, so these are $c(f-1)$ constraints.

So in total, the independent variables are $2 + c - f$ and this number must be positive, so the Gibbs rule states that

$$2 + c - f \geq 0$$

Statistical Mechanics

14 Statistical approach to TD

Thermodynamic systems contain very large numbers of particles, $N \sim \mathcal{N}_A \sim 10^{23}$, so the possible states are characterized by a large number of quantum numbers/canonical coordinates ($6N$ for point-like classic particles).
Example For quantum harmonic oscillators the microstates can be determined by the 3 quantum number for every oscillator, so a generic state is

$$|\nu(t)\rangle = \sum_A C_A(t) \, |\mathbf{n}_1, \mathbf{n}_2, ...\mathbf{n}_N\rangle \, .$$

Microstates evolve according to microscopic dynamics: Hamilton's equations

$$\begin{cases} \dot{q}^A = \dfrac{\partial H}{\partial p_A} \\ \dot{p}_A = -\dfrac{\partial H}{\partial q^A} \end{cases}$$

are $3N$ equations that, along with $6N$ initial conditions, completely determine the motion of the system.
Observation: the classical evolution is at odds with thermodynamics for 2 reasons:

1. Hamilton's equations obey the time-reversal symmetry, while the second principle of thermodynamics prescribes a preferential direction of the evolution of the system;
2. Poincaré cycles states that, if a phase space is limited, the trajectory of the system will return arbitrarily close to the starting point, given enough time.

This microscopic approach is not only inconceivable, since it involves solving $\sim \mathcal{N}_A$ coupled PDEs, but also redundant, as in TD the state of a gas could be determined with only a handful of macroscopic variables. The set of these variables is called a **macrostate** and each one includes a huge number of microstates: this state is at equilibrium, but the system still moves in the phase space and "visits" many microstates of the macrostate.

15 Observables

To measure an observable, the whole microstate must be known and the observable must be measured instantaneously: the value of the observable is the mean of the values found

$$O_{\text{obs}} = \frac{1}{M} \sum_{n=1}^{M} O_n$$

and the mean value in time is the integral mean over time on the system trajectory $\nu(t)$

$$\bar{O} = \frac{1}{T} \int_0^T \mathrm{d}t O(\nu(t)).$$

The statistical approach is different, the value is found using the average in terms of frequencies: given the set of microstates $\{\nu\}$, the value is

$$O_{\text{obs}} = \sum_{\{\nu\}} \frac{n(\nu)}{M} O(\nu)$$

where $n(\nu)$ is the number of times ????

This is similar to the definition of a mean value, with $P(\nu) = \frac{n(\nu)}{M}$: since the microstates are points in the phase space, the probability should be expressed in terms of an integral

$$\langle O \rangle = \int \mathrm{d}\mu \rho(q,p) O(q,p) = \int \mathrm{d}q \wedge \mathrm{d}p \rho(q,p) O(q,p)$$

where $\rho(q,p)$ is the probability that the microstate falls into an (hyper-)cube around (q,p); the main objective of statistical mechanics is to determine this probability density.

In general, for non-equilibrium states, the density evolves with time, but this evolution is determined by the *Liouville Theorem*.

Liouville Theorem

If a probability density $\rho(q,p,t)$ satisfies a continuity equation, then

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \frac{\partial \rho}{\partial t} + \sum_A \left(\frac{\partial \rho}{\partial q^A} \dot{q}_A + \frac{\partial \rho}{\partial p_A} \dot{p}_A \right) = \frac{\partial \rho}{\partial t} + \{\rho, H\} = 0$$

Proof

The continuity equation is

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{j} = 0$$

and the associated current $\mathbf{j}$ is given by "density times velocity", which in the phase space is

$$\mathbf{j} = \left(\rho \dot{q}^A, \rho \dot{p}_A \right)$$

expanding the divergence

$$0 = \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial q^A} \left(\rho \dot{q}^A \right) + \frac{\partial}{\partial p_A} \left(\rho \dot{p}_A \right) = \frac{\partial \rho}{\partial t} + \frac{\partial \rho}{\partial q^A} \dot{q}^A + \frac{\partial \rho}{\partial p_A} \dot{p}_A + \rho \left(\frac{\partial \dot{q}^A}{\partial q^A} + \frac{\partial \dot{p}_A}{\partial p_A} \right)$$

recalling Hamilton's equations and commuting second derivatives yields the result.

Lesson 6, 13/10

In this formalism, the equilibrium is found when the explicit dependency of ρ on time is null:

$$\frac{\partial \rho}{\partial t} = 0 \iff \{\rho, H\} = 0$$

which means that ρ should be a *constant of motion*, depending on q, p and other constants of motion.

16 Description of TD systems through independent subsystems

Consider a TD system composed of quasi-isolated subsystems (i.e. the interaction between subsystems are confined to their boundaries): for this reason, their probability densities can be well approximated as independent, which means

$$\rho_{ij}(q_i, p_i, q_j, p_j) = \rho_i(q_i, p_i) \rho_j(q_j, p_j)$$

for every subsystems $i \neq j$.

16.1 Additive observables

These observables can be expressed as the sum of the observable in each subsystem

$$O = \sum_{i=1}^{n} O_i,$$

like the total internal energy; their expectation values are also additive.

For this approach to be useful, the inevitable fluctuations about the equilibrium state must be small: consider the variance

$$\sigma^2 = \langle O^2 \rangle - \langle O \rangle^2 = \langle \sum_{ij} O_i O_j \rangle - \langle \sum_{i} O_i \rangle^2 = \sum_{ij} \left(\langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle \right) =$$

$$= \sum_{i} \left(\langle O_i^2 \rangle - \langle O_i \rangle^2 \right) + \sum_{i \neq j} \left(\langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle \right) = \sum_{i} \sigma_i^2$$

where the last sum vanishes for the quasi-isolation condition.

The mean value scales with n, while the standard deviation scales like $\sqrt{n}$: their quotient becomes smaller as n gets bigger, so the statistical approach becomes better the bigger the number of subsystems.

16.2 Explicit formula for ρ

Since $\rho_{ij} = \rho_i \rho_j$, $\ln(\rho_{ij}) = \ln(\rho_i) + \ln(\rho_j)$ is an extensive quantity and so should only depend on *additive* constants of motion.

From classical mechanics it is known that there exist **only 7 additive constants of motion**:

$$E_i,\ \mathbf{P}_i,\ \mathbf{L}_i$$

for the i'th subsystem; so the log of the probability density must be a function in the form

$$\ln(\rho_i) = \alpha + \beta E_i + \gamma \cdot \mathbf{P_i} + \delta \cdot \mathbf{L_i}$$

where the constants are independent of the subsystem and completely determined by the *total motion of the system*: the only non-trivial dependency of ρ at equilibrium is **energy**, because the total momentum and angular momentum can be factored out with a change of reference frame.

17 The Time reversal problem

One of the main problems in statistical mechanics is the fact that microscopic laws are time-reversible, while the macroscopic laws of TD are not.

17.1 Reduced Observables

A one-body observable is an observable in the form

$$O_1(q,p) = \sum_{i=1}^{N} O(\mathbf{q_i}, \mathbf{p_i})$$

for example $E_k = \sum \mathbf{p_i}^2 / 2m$.

A two-body observable is an observable in the form

$$O_2(q,p) = \sum_{i<j=1}^{N} O(\mathbf{q_i}, \mathbf{p_i}, \mathbf{q_j}, \mathbf{p_j}).$$

Consider the expected value of a 1-body observable

$$\langle O_1(t)\rangle = \int \mathrm{d}\mu_1 ... \mathrm{d}\mu_N \sum_{i=1}^{N} O(i)\rho(1,2,...N)$$

the density does not distinguish the particles, so ρ must be symmetric for variable swap: this means that the sum can be factored out as a N factor

$$\langle O_1(t)\rangle = N \int \mathrm{d}\mu_1 O(1) \underbrace{\int \mathrm{d}\mu_2 ... \mathrm{d}\mu_N \rho(1,2,...N)}_{:=\rho_1(1;t)}$$

this ρ_1 is called the **reduced probability density of order 1** and it is clearly normalized.

The same can be done for two-body observables

$$\langle O_2(t)\rangle = \int \mathrm{d}\mu_1 ... \mathrm{d}\mu_N \sum_{i<j} O(i,j)\rho(1,2...N)$$

The double sum is comprised of $\binom{N}{2}$ equal terms, so the **reduced probability of order 2** is

$$\binom{N}{2} \int \mathrm{d}\mu_1 \mathrm{d}\mu_2 O(1,2) \underbrace{\int \mathrm{d}\mu_3 ... \mathrm{d}\mu_N \rho(1,2,...N)}_{:=\rho_2(1,2;t)}$$

Assumptions (Diluted Gas):

- $H = \sum_i \frac{\mathbf{p}_i^2}{2m} + \sum_{i<j} \phi_{ij}$;
- $\phi_{ij} = \phi_{ji} = \phi(|\mathbf{q}_i - \mathbf{q_j}|)$
- "Short range" potential: $\phi_{ij} = 0$ for $|\mathbf{q}_i - \mathbf{q_j}| > r_0$

The Hamilton equations are

$$\begin{cases} \dfrac{\partial H}{\partial \mathbf{p}_i} = \dfrac{\mathbf{p}_i}{m} \\ \dfrac{\partial H}{\partial \mathbf{q}_i} = \displaystyle\sum_{k<l} \frac{\partial \phi_{kl}}{\partial \mathbf{q}_i} \end{cases}$$

the second equation can be simplified to

$$\frac{\partial H}{\partial \mathbf{q}_i} = \sum_{i<l} \frac{\partial \phi_{il}}{\partial \mathbf{q}_i} + \underbrace{\sum_{k<i} \frac{\partial \phi_{ki}}{\partial \mathbf{q}_i}}_{=\sum_{i<k} \cdots} = \sum_{j\neq i} \frac{\partial \phi_{ij}}{\partial \mathbf{q}_i} := -\mathbf{K}_{ij}$$

where $\mathbf{K}_{ij} = \mathbf{K}_{ji}$: Liouville's theorem becomes

$$\frac{\partial \rho}{\partial t} + \sum_i \frac{\mathbf{p}_i}{m} \frac{\partial \rho}{\partial \mathbf{q}_i} + \sum_{i\neq j} \mathbf{K_{ij}} \frac{\partial \rho}{\partial \mathbf{p}_i} = 0$$

antisymmetrizing the second sum yields

$$\frac{\partial \rho}{\partial t} + \sum_i \frac{\mathbf{p}_i}{m} \frac{\partial \rho}{\partial \mathbf{q}_i} + \frac{1}{2} \sum_{i\neq j} \mathbf{K_{ij}} \left(\frac{\partial \rho}{\partial \mathbf{p}_i} \frac{\partial \rho}{\partial \mathbf{p}_j} \right) = 0.$$

Defining the operators

$$\hat{S}_i = \frac{\mathbf{p}_i}{m}\frac{\partial}{\partial \mathbf{q}_i}$$
$$\hat{P}_{ij} = K_{ij}\left(\frac{\partial}{\partial \mathbf{p}_i} - \frac{\partial}{\partial \mathbf{p}_j}\right)$$

the equation becomes

$$(\frac{\partial}{\partial t} + \underbrace{\sum_{i=1}^{N}\hat{S}_i + \frac{1}{2}\sum_{i\neq j=1}^{N}\hat{P}_{ij}}_{:=\hat{L}_N})\rho = 0.$$

To find the consequences of this calculation on the reduced probability density, it is useful to redefine them with a different normalization:

$$f_1(i) = N\rho_1(i)$$
$$f_s(1,2,...s) = \frac{N!}{(N-S)!}\int \mathrm{d}\mu_{s+1}...\mathrm{d}\mu_N \rho(1,2...N) = \frac{N!}{(N-S)!}\rho_s(1,2...s)$$

where the second formula is the generalization of the reduced probability density at s bodies.

The relation with the Liouville operator $\hat{L}_N$ is

$$\frac{\partial f_s}{\partial t} = -\frac{N!}{(N-S)!}\int \mathrm{d}\mu_{s+1}...\mathrm{d}\mu_N \hat{L}_N \rho$$

It is better to subdivide $\hat{L}_N$ in the following fashion

$$\hat{L}_N = \sum_{i=1}^{s}\hat{S}_i + \sum_{j=s+1}^{N}\hat{S}_j + \frac{1}{2}\sum_{i\neq j=1}^{s}\hat{P}_{ij} + \frac{1}{2}\sum_{i\neq j=s+1}^{N}\hat{P}_{ij} + \sum_{i=1}^{s}\sum_{j=s+1}^{N}\hat{P}_{ij} = \hat{L}_s + \hat{L}_{N-s} + \sum_{i=1}^{s}\sum_{j=s+1}^{N}\hat{P}_{ij}$$

So the derivative becomes

$$\frac{\partial f_s}{\partial t}\frac{N!}{(N-s)!}\int \mathrm{d}\mu_{s+1}...\mathrm{d}\mu_N \left(\hat{L}_s\rho + \underbrace{\hat{L}_{N-s}\rho}_{} + \sum_{i=1}^{s}\sum_{j=s+1}^{N}\hat{P}_{ij}\rho\right)$$

the highlighted part is zero, because its parts can be factored and integrated by parts, e.g.

$$\int \mathrm{d}^3 p_{s+1}\mathrm{d}^3 q_{s+1}\frac{\mathbf{p}_{s+1}}{m}\frac{\partial \rho}{\partial \mathbf{q}_{s+1}} = [...?!]$$

and so reshuffling the terms

$$\left(\frac{\partial}{\partial t} + \hat{L}_s\right) f_s = -\frac{N!}{(N-s)!}\int \mathrm{d}\mu_{s+1}...\mathrm{d}\mu_N \sum_{i=1}^{s}\sum_{j=s+1}^{N}\hat{P}_{ij}\rho$$

since the $s+1,...,N$ coordinates are integrated, the sum over j gives the same result $N-s$ times, using $j = s+1$ yields

$$\left(\frac{\partial}{\partial t} + \hat{L}_s\right) f_s = -\frac{N!}{(N-s)!}(N-s)\sum_{i=1}^{s}\int \mathrm{d}\mu_{s+1}...\mathrm{d}\mu_N \hat{P}_{i,s+1}\rho =$$
$$= -\sum_{i=1}^{s}\int \mathrm{d}\mu_{s+1}\hat{P}_{i,s+1}\underbrace{\frac{N!}{(N-(s+1))!}\int \mathrm{d}\mu_{s+2}...\mathrm{d}\mu_N \rho}_{:=f_{s+1}} =$$

$$= -\sum_{i=1}^{s} \int \mathrm{d}\mu_{s+1} \mathbf{K}_{i,s+1} \left(\frac{\partial}{\partial \mathbf{p}_i} - \frac{\partial}{\partial \mathbf{p}_{s+1}} \right) f_{s+1}(1, 2..., s+1; t)$$

the derivative with respect to $\mathbf{p}_{s+1}$ can be cancelled integating by parts.

The final result is the BBGKY hierarchy

$$\left(\frac{\partial}{\partial t} + \hat{L}_s \right) f_s(1, 2...s; t) = -\sum_{i=1}^{s} \int \mathrm{d}\mu_{s+1} \mathbf{K}_{i,s+1} \frac{\partial}{\partial \mathbf{p}_i} f_{s+1}(1, 2, ..., s+1; t)$$

Lesson 7, 14/10

Is there some physical region to discard some equations?

17.2 Time scales in BBGKY

In the hierarchy, there are three main time scales

- $\mathbf{K} \cdot \frac{\partial}{\partial \mathbf{p}} \sim \frac{1}{\tau}$ collision time
- $\frac{\mathbf{p}}{m} \frac{\partial}{\partial \mathbf{q}} \sim \frac{1}{\tau_s}$ typical time in which the particle travels a space over which f changes significantly

It is true that

$$\tau << \tau_s$$

[??? did not follow]:
f_2 varies more than f_1, so the assumption is that it reaches the equilibrium state faster than f_1

[???? sto dormendo, chiedi a Remi il resto]

Lesson 8, 20/10

18 Boltzmann H-theorem

[smth about sufficient and necessary condition]

$$\mathbb{H}(t) = \int \mathrm{d}^3 v f(\mathbf{p}, t) \ln(f)$$

since f is taken to be homogeneous in space, it does not depend on $\mathbf{q}$.

Its derivative is

$$\frac{\mathrm{d}\mathbb{H}}{\mathrm{d}t} = \int \mathrm{d}^3 v \frac{\partial f}{\partial t} (\ln(f) + 1)$$

so

$$\frac{\partial f}{\partial t} = 0 \implies \frac{\mathrm{d}\mathbb{H}}{\mathrm{d}t} = 0.$$

Is it true that

$$\frac{\mathrm{d}\mathbb{H}}{\mathrm{d}t} = 0 \overset{?}{\iff} f(1')f(2') - f(1)f(2) = 0$$

when f satisfies the Boltzmann equation?

Consider the derivative of $\mathbb{H}$ just derived

$$\frac{\mathrm{d}\mathbb{H}}{\mathrm{d}t} = \int \mathrm{d}^3 v \frac{\partial f(1)}{\partial t} \left(1 + \ln(f(1))\right) = \int \frac{\mathrm{d}^3 p_1}{m} \left(\frac{\partial f_1}{\partial t}\right)_{\text{coll}} \left(1 + \ln(f_1)\right)$$

Applying Boltzmann equation yields

$$\frac{\mathrm{d}\mathbb{H}}{\mathrm{d}t} = \int \mathrm{d}^3 p_1 \frac{1}{m} \mathrm{d}^3 p_2 \mathrm{d}^3 p_1' \mathrm{d}^3 p_2' \delta^4(P_f - P_i) |T_{fi}|^2 \left(f(1')f(2') - f(1)f(2)\right) \left(1 + \ln(f(1))\right)$$

since T_{fi} is symmetric under exchange of 1 and 2, one can symmetrize this expression in

$$\frac{\mathrm{d}\mathbb{H}}{\mathrm{d}t} = \frac{1}{2m} \int \mathrm{d}^3 p_1 \mathrm{d}^3 p_2 \mathrm{d}^3 p_1' \mathrm{d}^3 p_2' \delta^4(P_f - P_i) |T_{fi}|^2 \left(f(1')f(2') - f(1)f(2)\right) \left(2 + \ln(f(1) + \ln(f(2)))\right)$$

but T_{fi} is also invariant under the exchange $(1, 2) \to (1', 2')$ and so holds

$$\frac{\mathrm{d}\mathbb{H}}{\mathrm{d}t} = \frac{1}{2m} \int \mathrm{d}^3 p_1 \mathrm{d}^3 p_2 \mathrm{d}^3 p_1' \mathrm{d}^3 p_2' \delta^4(P_f - P_i) |T_{fi}|^2 \left(f(1)f(2) - f(1')f(2')\right) \left(2 + \ln\big(f(1') + \ln\big(f(2')\big)\big)\right)$$

The average between these two equations can be taken to be

$$\frac{\mathrm{d}\mathbb{H}}{\mathrm{d}t} = \frac{1}{4m} \int \mathrm{d}^3 p_1 ... \delta^4(P_f - P_i) |T_{fi}|^2 \left(f(1')f(2') - f(1)f(2)\right) \left(\ln(f(1)f(2)) - \ln\big(f(1')f(2')\big)\right)$$

it can be shown that the last product is strictly negative:

$$h(x, y) = (y - x)(\ln x - \ln y), \text{ with } x, y \geq 0$$

$$h(x, y) = (y - x) \ln \frac{x}{y} = y \ln \frac{x}{y} \left(1 - \frac{x}{y}\right)$$

so if $\frac{x}{y} \geq 0$, then the ln is positive, while the difference is negative and vice versa if $\frac{x}{y} \leq 0$, so $h(x, y) < 0$.

The statement of the Boltzmann $H-$theorem is then

$$\frac{\mathrm{d}\mathbb{H}}{\mathrm{d}t} \leq 0$$

with

$$\frac{\mathrm{d}\mathbb{H}}{\mathrm{d}t} = 0 \iff f(1')f(2') - f(1)f(2).$$

With some assumptions [e.g. molecular chaos, last lecture i think]

$$\frac{\partial f}{\partial t} = 0 \iff f(1')f(2') - f(1)f(2) = 0$$

19 Maxwell-Boltzmann Distribution

At equilibrium

$$f(\mathbf{p_1}) f(\mathbf{p_2}) = f(\mathbf{p_1'}) f(\mathbf{p_2'})$$

when energy and momentum are conserved, i.e.

$$\begin{cases} \mathbf{p_1} + \mathbf{p_2} = \mathbf{p_1'} + \mathbf{p_2'} \\ |\mathbf{p_1}|^2 + |\mathbf{p_2}|^2 = |\mathbf{p_1'}|^2 + |\mathbf{p_2'}|^2 \end{cases}$$

The ln of the equilibrium condition yields

$$\ln f(\mathbf{p_1}) + \ln f(\mathbf{p_2}) = \ln f(\mathbf{p_1'}) + \ln f(\mathbf{p_2'})$$

these conditions, as functional relations, imply that the ln is in the form

$$\ln f(\mathbf{p}) = a + \mathbf{b}\cdot\mathbf{p} - c|p|^2$$

the b dependency can be taken out by considering a frame in which $\langle p\rangle = 0$ (check it! Complete the Gaussian Integral.), so the function is in the form

$$f(\mathbf{p}) = Ae^{-c|\mathbf{p}|^2}.$$

Its normalization must be

$$\int \mathrm{d}^3 p f(\mathbf{p}) = \frac{N}{V};$$

in addition to this, the expression $E = \frac{3}{2}NkT$ has to be interpreted as

$$\langle \frac{|\mathbf{p}|^2}{2m}\rangle = \frac{3}{2}kT$$

. Using spherical coordinates, since $f(\mathbf{p}) = f(p)$, yields

$$\frac{N}{V} = 4\pi\int_0^\infty \mathrm{d}p p^2 Ae^{-cp^2}$$
$$\frac{3}{2}kT\frac{N}{V} = 4\pi\int_0^\infty \mathrm{d}p p^2 \frac{p^2}{2m}Ae^{-cp^2}$$

Using the general formula

$$\int_0^\infty \mathrm{d}x x^{2k}e^{-cx^2} = \frac{1}{2}\frac{\Gamma\left(k+\frac{1}{2}\right)}{c^{k+1/2}}$$

yields

$$\frac{N}{V} = 4\pi A\frac{\sqrt{\pi}}{4c^{3/2}}$$
$$\frac{3kTN}{2V} = \frac{4\pi A}{2m}\frac{3\sqrt{\pi}}{8c^{5/2}}$$

Taking the quotient of the expressions yields

$$\frac{1}{2mc} = kT \iff c = \frac{1}{2mkT}$$

and also

$$A = \frac{N}{V}\left(\frac{1}{2\pi mkT}\right)^{\frac{3}{2}}$$

So the Maxwell Boltzmann distribution is

$$f_{MB}(p) = \frac{N}{V}\left(\frac{1}{2\pi mkT}\right)^{\frac{3}{2}}\exp\left[-\frac{p^2}{2m}\frac{1}{kT}\right]$$

19.1 About the H-theorem

There exist $\mathbb{H}$ so that its derivative is non-positive.

- Is $\mathbb{H} \propto -S$? The expression of $\mathbb{H}$ is similar to Gibbs' approach to the entropy
- ...other things [time reversal etc.]

If at time t molecular chaos holds out of equilibrium, i.e. $\frac{\mathrm{d}\mathbb{H}}{\mathrm{d}t} < 0$, at time $t - \varepsilon$ this derivative is positive! [did not understand...]

20 Going Quantum

In general, classical systems are just approximations of quantum systems: the state of the system is in general a vector in an Hilbert space

$$|\psi\rangle \in \mathcal{H}$$

and can be expressed in a convenient basis, e.g. position basis

$$|\psi(t)\rangle = \int \mathrm{d}q\, |q\rangle \underbrace{\langle q|\psi(t)\rangle}_{:=\psi(q,t)}.$$

Time evolution is dictated by Schrödinger's equation

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$$

in which time-reversibility is always a problem.

Consider the evolution of a *pure* state, defined by the eigenvalues $\{n\}$ of a complete set of observables, including the Hamiltonian: for a time independent Hamiltonian it holds

$$\hat{H} |n(t)\rangle = E_n |n(t)\rangle$$

with

$$|n(t)\rangle = e^{-\frac{i}{\hbar}E_n t}$$

and the generic pure state can be expressed in terms of these stationary states

$$|\psi(t)\rangle = \sum\!\!\!\!\!\!\int c_n e^{-\frac{i}{\hbar}E_n t} |n\rangle$$

The expectation value of an observable $\hat{A}$ is

$$\langle \hat{A} \rangle_\psi = \langle \psi(t)| \hat{A} |\psi(t)\rangle = \sum_n |c_n|^2 \langle n| \hat{A} |n\rangle + \sum_{m \neq n} c_n c_m^* e^{\frac{i}{\hbar}(E_m - E_n)t} \langle m| \hat{A} |n\rangle$$

From this expression it seems like isolated quantum *do not* thermalize, but there has been recent progress, called Eigenvalue Thermalization Hypothesis: for *some* observables with some randomness in the interactions, it has been proven that this happens.

20.1 Density matrix

The initial state cannot be completely known, real systems are not isolated, but quasi-isolated at best and every interaction modifies the energy levels.

To solve the first problem, the **density matrix** is used, using the classical probabilities $0 \leq p_k \leq 1$ of the system being in the state $|\psi_k\rangle$ (which is not guaranteed to be an ONC basis):

$$\hat{\rho} = \sum_k p_k |\psi_k\rangle \langle \psi_k|$$

the value of the observables is

$$\langle \hat{A} \rangle = \sum_k p_k \langle \psi_k| \hat{A} |\psi_k\rangle = \mathrm{Tr}\Big(\hat{\rho}\hat{A}\Big)$$

Consider an ONC basis $\{|\alpha\rangle\}$:

$$\langle \hat{A} \rangle = \sum_{k,\alpha} p_k \langle \psi_k|\alpha\rangle \langle \alpha| \hat{A} |\psi_k\rangle = \sum_{k,\alpha} \langle \alpha| \Big(\hat{A} p_k |\psi_k\rangle \langle \psi_k|\Big) |\alpha\rangle = \mathrm{Tr}\Big(\hat{\rho}\hat{A}\Big)$$

20.1.1 Properties

- $\mathrm{Tr}(\hat{\rho}) = 1$;
- In any basis, the elements on the diagonal of ρ are

$$\rho_{\alpha\alpha} = \langle\alpha|\,\hat{\rho}\,|\alpha\rangle = \sum_k p_k |\,\langle\alpha|\psi_k\rangle\,|^2 \in [0,1]\,;$$

- Considering the basis that diagonalizes ρ yields the fact that

$$\mathrm{Tr}\big(\rho^2\big) \leq 1;$$

- A pure state is represented by the density matrix

$$\rho = |\psi\rangle\,\langle\psi| \iff \rho^2 = \rho \implies \mathrm{Tr}\big(\rho^2\big) = 1.$$

Lesson 9, 27/10

Since the off-diagonal elements represent transition apmlitudes, at equilibrium the matrix should be symmetric, i.e.

$$\rho_{\alpha\beta} = \rho_{\beta\alpha}.$$

The deistic matrix ρ has the same role as $\rho(p,q)$ in the classical pictures and obeys the equation

$$i\hbar\frac{\partial\rho}{\partial t} = [H,\rho] \to \rho(t) = e^{-\frac{i}{\hbar}Ht}\rho(0)e^{\frac{i}{\hbar}Ht}$$

Again, at equilibrium

$$\frac{\partial\rho}{\partial t} = o \iff [H,\rho] = 0$$

which means that ρ is a function only of operators that commute with H: the assumption taken is

$$\rho = \rho(H)$$

and at most of the number operator $\hat{N}$ if the system is open.

At equilibrium a convenient base is that of the stationary states of H

$$[\rho,H] = 0 \to \rho = \sum_n \rho_n\,|n\rangle\,\langle n|$$

Recall that the Von Neumann Entropy is defined as

$$S = -\,\mathrm{Tr}(\rho\ln(\rho)) = -\sum_i \lambda_i \ln\lambda_i$$

which measures the information stored in a state: $S = 0$ for pure states, $S > 0$ for mixed states.

21 Gibbs Statistical Ensembles

Recalling that in classical SM

$$O_{\mathrm{obs}} = \sum_\nu P(\nu)O(\nu) = \langle O\rangle$$

where ν is the set of microstates of the system compatible with the chosen macrostate. The goal of this part is to

- Determine the general form of $P(\nu)$ for isolated, closed... systems.
- Express TD quantities in terms of $P(\nu)$.

21.1 Microcanonical Ensemble

It describes **isolated systems at fixed $\mathbf{V}, \mathbf{N}, \mathbf{E}$**: note that there can be uncertainty on the value of E, but under the condition

$$\frac{\Delta E}{E} << 1.$$

At equilibrium, $P(\nu) = f(E)$ is constant, because $E(\nu)$ is, so recalling

$$\sum_{\nu=1}^{\Omega} P(\nu) = P \sum_{\nu=1}^{\Omega} = 1 \iff P \equiv \frac{1}{\Omega}$$

where Ω is the total number of microstates defined by E,V,N: this can be interpreted as the fact that the system explored all microstate in the nonzero time of the macroscopic measurement; this is called *ergodicity*.

Although it is realized for almost all macroscopic systems, it is problematic to establish rigorously [qualcosa su KAM theorem] and is surprisingly difficult to achieve. $P(\nu) = \frac{1}{\Omega}$ is admissible if and only if the return time of the system (i.e. the time for the system to return arbitrarily close to its initial position) is comparable to the measurement time: the problem is that the return time is typically very large. [qualcosa] the Kac Lemma

$$\tau_{\text{ret}} \propto \tau e^{\alpha N}$$

if $P(\nu) \propto e^{-\alpha N}$. [small things]

To find the TD quantities' expressions, $S(E, V, N)$ is needed: consider a system subdivided into 2 isolated parts $1, 2$; then the subdivision becomes so that they can exchange only heat: the total energy is the same, but the equilibrium variables vary. The number of microstates are

$$\Omega_1(E_1, V_1, N_1),\ \Omega_2(E_2, V_2, N_2) \to \Omega(E) = \Omega_1 \Omega_2$$

Before (?) the second equilibrium, the situation is

$$\Omega(E) = \int \mathrm{d}E_1 \Omega_1(E_1)\Omega_2(E - E_1)$$

then it reaches a situation in which the number of microstates is maximum, with energies $E_1 = \bar{E}_1,\ E_2 = \bar{E}_2$.

Ω_1 at equilibrium must be sharply peaked at the value $\bar{E}_1$, which maximises the integrand:

$$0 = \frac{\partial}{\partial E_1}\left(\Omega_1(E_1)\Omega_2(E - E_1)\right) = \frac{\partial \Omega_1}{\partial E_1}\Omega_2(E - E_1) - \Omega_1(E_1)\frac{\partial \Omega_2}{\partial E_2}$$

$$\frac{1}{\Omega_1(E_1)}\frac{\partial \Omega_1}{\partial E_1} = \frac{1}{\Omega_2(E_2)}\frac{\partial \Omega_2}{\partial E_2}$$

$$\frac{\partial \ln \Omega_1}{\partial E_1} = \frac{\partial \ln \Omega_2}{\partial E_2}$$

since the TD equilibrium condition is $T_1 = T_2$, which means

$$\frac{1}{T_1} = \frac{1}{T_2} \iff \left.\frac{\partial S}{\partial E_1}\right|_{V_1, N_1} = \left.\frac{\partial S}{\partial E_2}\right|_{V_2, N_2}$$

comparing the two expressions yields

$$S_i \propto \ln \Omega_i \to S = k \ln \Omega$$

where k can be experimentally proven to be a universal constant, the Boltzmann constant

$$k = \frac{R}{N_A} = 1.38 \cdot 10^{-23}\ \mathrm{JK}^{-1}.$$

Since this arose from a confront of derivatives, a constant offset could have been added, but $\Omega = 1 \to S = 0$ is convenient, since Nernst's principle prescribes that at $T = 0$ the entropy must go to 0.

Note This form of S preserves the fact that it is an extensive quantity.

The general expression for entropy will be

$$S = -k \sum_{\nu} P(\nu) \ln P(\nu)$$

21.2 Example: Classical ideal gas

The Hamiltonian is the free N particle Hamiltonian

$$H = \frac{1}{2m} \sum_{i=1}^{N} p_i^2$$

and the microcanonical ensemble is a region of the phase space such as

$$E \leq H \leq E + \Delta E, \ \frac{\Delta E}{E} << 1$$

the possible values of the momenta live in a shell of width ΔE around the hypersurface

$$\sum_{i=1}^{N} p_i^2 = 2mE := R^2(E)$$

which is a $(3N-1)$-sphere, so the volume of this region is, at first order, the area of this sphere multiplied by its width:

$$V \approx \Sigma_{3N+1}(R(E)) \cdot (R(E + \Delta E) - R(E))$$

Note The surface of a unit D-sphere.

Consider the integral

$$I_D = \int \mathrm{d}^D x e^{-(\mathbf{x})^2} = \left(\int \mathrm{d}x e^{-x^2} \right)^D = (\pi)^{D/2}$$

this can also be computed in (hyper-)spherical coordinates

$$I_D = \int \mathrm{d}\Omega \int \mathrm{d}r r^{D-1} e^{-r^2}$$

the integral over the hyper-solid angle Ω is the area of the $(D-1)$-sphere! So

$$I_D = \Sigma_{D-1} \int_0^{\infty} \mathrm{d}r r^{D-1} e^{-r^2}$$

letting $t = r^2 \to \mathrm{d}t = 2r\mathrm{d}r$ yields

$$I_D = \Sigma_{D-1} \frac{1}{2} \int \mathrm{d}t t^{D/2-1} e^{-t} = \Sigma_{D-1} \frac{\Gamma(D/2)}{2}$$

so

$$\Sigma_{D-1} = \frac{2\pi^{D/2}}{\Gamma(D/2)}$$

Lesson 10, 28/10

The width of the hyper-spherical surface can also be expressed as

$$\frac{\partial R}{\partial E}\Delta E$$

so the volume in the momentum space is

$$V_p = \Sigma_{3N-1} R^{3N-1}(E)\frac{\partial R}{\partial E}\Delta E.$$

The volume in the coordinates' space is

$$V_q = V^N$$

where V is the physical volume of the system. Recalling $R = \sqrt{2mE}$, the total volume in the phase space is

$$V = V^N \cdot V_p = [...]$$

To get a number of microstates out of this volume, a measure must be defined, i.e. a "unit volume", that can has to have the dimension of an action to the power of $3N$, one can use Planck's constant

$$h^{3N}$$

and the number of microstates becomes

$$\Omega(E, V, N) = \frac{V(E, V, N)}{h^{3N}} :$$

this is problematic, as the entropy derived from this expression is **not extensive**, so a factor of $N!$ must be *forcibly introduced* in the denominator; this is **not justified** in a classical statistical theory, as is the choice of h, but these assumptions make sense in a quantum framework: the $N!$ is due to *indistinguishability* of the particles and the h^{3N} is due to the fact that $\Delta q \Delta p \sim h$, so this is a sensible choice.

$$\Omega(E, V, N) = \frac{V(E, V, N)}{h^{3N} N!} = \left(\frac{2\pi m V^{\frac{2}{3}} E}{h^2}\right)^{\frac{3N}{2}} \frac{1}{N!\Gamma\left(\frac{3N}{2}\right)}\frac{\Delta E}{E}.$$

Now Boltzmann's formula can be used to derive the entropy

$$S(E, V, N) = k \ln \Omega(E, V, N) :$$

recalling Stirling's approximation

$$\ln N! = N \ln N - N + ...$$
$$\ln \Gamma(M) = M \ln M - M + ...$$

that holds in the TD limit ($N, V, E \to \infty$), one gets

$$S = k\left\{\ln\left(\frac{2\pi m V^{\frac{2}{3}} E}{h^2}\right)^{\frac{3N}{2}} - \ln N! - \ln\Gamma\left(\frac{3N}{2}\right) + \ln\frac{\Delta E}{E}\right\}$$

The last term is independent of N, so it can be discarded in the TD limit. Some algebra shows that

$$S(E, V, N) = kN \ln\left[\frac{V}{N}\left(\frac{4\pi m E}{3h^2 N}\right)^{\frac{3}{2}}\right] + \frac{5}{2}kN$$

this is the Sakur-Tetrode entropy.

The $N!$ factor produces the $\frac{1}{N}$ factor inside the ln: without it, since the volume grows as N, the entropy would grow as $S \sim N \ln N$, which is not the case in the physical world.

In the microcanonical ensemble, the thermal equation of state is

$$\frac{1}{T} = \left.\frac{\partial S}{\partial E}\right|_{V,N} = kN\frac{\partial \ln E^{\frac{3}{2}}}{\partial E} \implies E = \frac{3}{2}kNT$$

since in thermodynamics

$$E = \frac{3}{2}nRT$$

Boltzmann's constant has value

$$k = \frac{R}{N_A}.$$

Note There is no reason to fix the value of k a priori, so every confront with TD must yield this value of k for it to be a universal constant.

The mechanical equation of state is

$$\frac{p}{T} = \left.\frac{\partial S}{\partial V}\right|_{E,N} = kN\ln\frac{V}{N} \sim \frac{kN}{V} \implies pV = kNT$$

which is the ideal gas equation of state.

As an exercise, derive the chemical equation of state

$$\frac{\mu}{T} = \left.\frac{\partial S}{\partial N}\right|_{V,E}.$$

Using the last two results

$$pV = \frac{2}{3}E$$

where the $\frac{2}{3}$ come from the fact that in the expression of Ω the only adimensional combination of m, V, E, h is the one provided.

21.3 Example: Quantum Mechanics

Since the microcanonical ensemble is used, all equilibrium microstates are equiprobable.

A set of stationary states $\{|n\rangle\}$ with $n = 1, 2...\Omega$ and with eigenvalues $E_n \in (E, E + \Delta E)$.

As discussed before, the density matrix at equilibrium must be diagonal in the eigenstates' basis

$$\rho_{mn} = \frac{1}{\Omega}\delta_{mn} \to \rho = \frac{1}{\Omega}\sum_{n=1}^{\Omega}|n\rangle\langle n|$$

The corresponding Von Neumann's entropy

$$S = -\operatorname{Tr}(\rho\ln\rho) = -\sum_n \lambda_n \ln\lambda_n = -\sum_{n=1}^{\Omega}\frac{1}{\Omega}\ln\frac{1}{\Omega} = \ln\Omega$$

The postulate is that the TD entropy is proportional to Von Neumann's:

$$S_B = kS_{\text{VN}} \to S = k\ln\Omega$$

Note Since the eigenvalues are constant, for the microcanonical ensemble the density matrix is diagonal in *all bases*, but this is not generally true for other ensembles.

In a QM context, the ensemble is an incoherent superposition of states, with random phases [altre cose]

$$\rho = \frac{1}{\Omega}\sum_{\alpha} |\alpha\rangle \langle\alpha| \,, \; |\alpha\rangle = \sum_n c_{\alpha,n} |n\rangle = \sum_n |c_{\alpha,n}| e^{i\vartheta_{\alpha,n}}$$

the randomness of the phases means

$$\langle e^{i\vartheta_{\alpha,m}} e^{i\vartheta_{\alpha,n}} \rangle = 0$$

uniting these three relations with the fact that

$$|c_{\alpha,n}| = \text{const.}$$

shows that this last condition ensures

$$\rho_{mn} = 0, \text{ if } m \neq n$$

.

The explicit quantum example is the *Einstein Solid*: the thermal capacity c_v is classically constant (Dulong-Petit), while experiments show that this is not true for low temperatures.

To solve this, quantum mechanics is needed and Einstein proposed a simple quantum model: the atoms of a solid live in a 3D lattice in fixed positions, about which they **oscillate**, so a solid stores energy in these oscillations; Einstein considered a simplified model, in which only one frequency is permitted, to be treated quantum mechanically: so the situation is that of $3N$ decoupled quantum harmonic oscillators with only one permitted frequency.

Each stationary state (each corresponding to one oscillator) is defined by a quantum number n_a, with $a = 1, 2...3N$, so the generic state of the system is

$$|n\rangle = |n_1\rangle \otimes |n_2\rangle \otimes ... |n_{3N}\rangle$$

The total energy of the system is the sum of the oscillators' energy

$$E = \sum_{a=1}^{3N} E_{n_a} = \hbar\omega \left(\underbrace{\sum_{a=1}^{N} n_a}_{:=n} + \frac{3N}{2} \right).$$

Consider the microcanonical ensemble, i.e. the set of states with energy $E = \hbar\omega \left(n + \frac{3N}{2}\right)$ and with $\Delta E = 0$, since the discrete spectrum allows it. Now, finding the number of microstates Ω is a *combinatorics* problem, since all the combinations of $\{n_a\}$ so that $n = \sum_a n_a$ must be counted, i.e. the possible positions of the quanta of energy in the $3N$ oscillators.

This is simply a distribution problem: how many ways are there to distribute n identical objects in $3N$ groups?

$$\Omega(n, N) = \binom{n + 3N - 1}{n}$$

from this the process is the same as before:

$$S = k \ln \Omega$$

then Stirling's approximation, with the caveat $3N - 1 \sim 3N$, yields

$$S = k \left\{ (n + 3N) \ln(n + 3N) - n \ln N - 3N \ln 3N \right\}.$$

Then

$$\frac{1}{T} = \frac{\partial S}{\partial E} = \frac{\partial n}{\partial E}\frac{\partial S}{\partial n} = \frac{k}{\hbar\omega} \left(\ln(n + 3N) + 1 - \ln(n) - 1 \right) = \frac{k}{\hbar\omega} \ln \frac{n + 3N}{n}.$$

Defining

$$\beta = \frac{1}{kT} \to \beta\hbar\omega = \ln\frac{n+3N}{n}$$

so

$$\frac{n+3N}{n} = e^{\beta\hbar\omega} \to n = \frac{3N}{1-e^{\beta\hbar\omega}}$$

since $n(E)$ and $\beta(T)$,

$$E(T,N) = \hbar\omega 3N\left(\frac{1}{e^{\beta\hbar\omega}-1}+\frac{1}{2}\right)$$

The quantum effects vanish if $kT >> \hbar\omega$, indeed

$$E \sim \hbar\omega 3N\left(\frac{1}{\beta\hbar\omega}+\frac{1}{2}\right) \sim \hbar\omega 3N\frac{1}{\beta\hbar\omega} = 3NkT$$

and Dulong-Petit's law appears:

$$c_v = \frac{\partial E}{\partial T} = 3Nk.$$

However, in a quantum regime, i.e. $kT << \hbar\omega$, so

$$E \sim \hbar\omega 3N\left(e^{-\beta\hbar\omega}+\frac{1}{2}\right) = \hbar\omega\frac{3}{2}N + \hbar\omega 3Ne^{-\beta\hbar\omega}$$

and

$$c_v = \frac{\partial E}{\partial T} \sim 3N\frac{(\hbar\omega)^2}{kT^2}e^{-\frac{\hbar\omega}{kT}}$$

which is better than a constant, but not in really good accord with the data.